Calcurion Maths Worksheets

GCSE Mathematics

Topic: Indices & Surds

Medium & Hard • 25 Questions

Scan for free resources

Section A – Medium (Q1–15)

1. Simplify $3^2 \times 3^4$. **[2]**

2. Simplify $\dfrac{5^7}{5^3}$. **[2]**

3. Simplify $(2^3)^4$. **[2]**

4. Simplify 4^{-2}. **[2]**

5. Simplify $8^{2/3}$. **[2]**

6. Simplify $27^{2/3}$. **[2]**

7. Simplify $\sqrt{50}$. **[2]**

8. Simplify $\sqrt{72}$. **[2]**

9. Simplify $\sqrt{12} + \sqrt{27}$. **[3]**

10. Simplify $\sqrt{50} - \sqrt{18}$. **[3]**

11. Rationalise $\dfrac{1}{\sqrt{5}}$. **[2]**

12. Rationalise $\dfrac{3}{\sqrt{7}}$. **[2]**

13. Rationalise $\dfrac{2}{\sqrt{3}-1}$. **[3]**

14. Expand and simplify $(\sqrt{3}+2)(\sqrt{3}-1)$. **[3]**

15. Expand and simplify $(2\sqrt{5}-1)^2$. **[3]**

Section B – Hard (Q16–25)

16. Simplify $\dfrac{16^{3/4}}{2^{-1}}$. **[3]**

17. Simplify $\sqrt{48}+2\sqrt{27}-\sqrt{12}$. **[4]**

18. Rationalise $\dfrac{\sqrt{3}}{\sqrt{5}-\sqrt{2}}$. **[4]**

19. Simplify $\sqrt{12}\times\sqrt{27}$. **[3]**

20. Express $\dfrac{1}{2+\sqrt{3}}$ in the form $a+b\sqrt{3}$. **[3]**

21. Express $\dfrac{5}{\sqrt{5}+2}$ in the form $a\sqrt{5}+b$. **[4]**

22. Solve $2^x = 32$. **[2]**

23. Solve $3^{2x} = 81$. **[3]**

24. Solve $4^x = 8^{x-1}$. **[4]**

25. Simplify $\left(\frac{8}{27}\right)^{-2/3}$. **[4]**

End of Worksheet • *Show all working clearly.*